Beyond Visual Quality: Evaluating the Robustness and Recognition Performance of Image Cartoonization Methods

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Abstract

Image cartoonization transforms photographs into stylized images, but visual appeal alone does not establish whether transformed images retain information useful for machine recognition. This study compares three pipelines —OpenCV-based cartoonization, pretrained AnimeGANv2, and a pretrained Stable Diffusion image-to-image pipeline—using a fixed ResNet18 classifier. A balanced subset of 300 images from ten Imagenette categories was evaluated on clean inputs and under Gaussian blur, Gaussian noise, low illumination, and JPEG compression. Recognition accuracy, structural similarity (SSIM), peak signal-to-noise ratio (PSNR), prediction consistency, per-class accuracy, confidence intervals, paired comparisons, confusion matrices, and failure-case examples were examined. On the common matched clean subset (n=294), original inputs achieved 80.61% accuracy, compared with 59.18% for AnimeGANv2, 33.33% for Stable Diffusion among its 294 valid outputs, and 18.03% for OpenCV. On matched samples under Gaussian blur (n=292), Gaussian noise (n=297), low illumination (n=295), and JPEG compression (n=294), AnimeGANv2 achieved 40.07%, 60.94%, 47.46%, and 57.82% accuracy, respectively, outperforming the other cartoonization methods in each condition. Its mean clean-output SSIM and PSNR were 0.6564 and 18.5636 dB. Six of 300 clean Stable Diffusion outputs were completely black (2.0% output failure); recognition accuracy conditional on valid outputs was 33.33%. Findings are limited to this sample, classifier, and implementation and require external validation.

Keywords: image cartoonization; image-to-image translation; AnimeGANv2; Stable Diffusion; ResNet18; robustness; image classification; SSIM; PSNR

1. Introduction

Image cartoonization is an image-to-image transformation task that changes photographs through texture simplification, contour emphasis, and altered color distributions. Research in image translation has produced conditional adversarial methods such as pix2pix and unpaired translation approaches such as CycleGAN, while dedicated cartoonization methods explicitly target cartoon-like surface, structure, and texture characteristics [1–4]. Latent diffusion models provide another generative paradigm for image synthesis and conditional editing [5].

A transformed image may look plausible while losing, exaggerating, or changing features used by a recognition model to distinguish object categories. Separately, robustness research has shown that common corruptions such as blur and noise can affect image-classifier performance, motivating controlled evaluation under image degradation [6]. This study therefore considers downstream recognition alongside reference-image similarity.

The study compares OpenCV cartoonization, pretrained AnimeGANv2, and a pretrained Stable Diffusion image-to-image pipeline using a fixed ResNet18 classifier. It evaluates clean images and four degradation conditions and examines recognition accuracy, SSIM, PSNR, prediction consistency, per-class results, paired comparisons, confusion matrices, and failure cases.

2. Related Work

2.1 Image-to-image translation and cartoonization

Isola et al. introduced pix2pix, a conditional adversarial framework for paired image-to-image translation [1]. Zhu et al. proposed CycleGAN, using cycle consistency to learn mappings between domains without paired examples [2]. These are general translation frameworks rather than methods designed solely for cartoonization.

Chen et al. proposed CartoonGAN, using semantic content loss and an edge-promoting adversarial loss to address the simplified textures and clear contours characteristic of cartoons [3]. Wang and Yu proposed learning to cartoonize with white-box cartoon representations, separating surface, structure, and texture representations [4]. These works show that content structure and edge preservation are important considerations in cartoonization.

2.2 Diffusion-based image synthesis

Rombach et al. introduced latent diffusion models, performing diffusion in a learned latent space to reduce computational demands while supporting high-quality image synthesis and conditioning [5]. Here, Stable Diffusion is treated as a pretrained image-to-image pipeline rather than a model trained specifically for cartoonization. Reproduction requires recording the exact model identifier/revision, prompt, seed, scheduler, inference steps, strength, guidance scale, and preprocessing.

2.3 Robustness and image-quality evaluation

Hendrycks and Dietterich introduced ImageNet-C and ImageNet-P benchmarks to evaluate classifier robustness to common corruptions and perturbations [6]. This study uses four selected corruptions—Gaussian blur, Gaussian noise, low illumination, and JPEG compression—not the full ImageNet-C benchmark.

Wang et al. introduced SSIM as a measure of structural information loss between a reference image and a distorted image [7]. PSNR is a pixel-error-based measure. These metrics describe similarity to the original image, not subjective artistic quality; stylization is expected to change pixels and structure, so metrics should be interpreted alongside recognition and visual inspection.

2.4 Recognition model and dataset

ResNet, introduced by He et al., uses residual learning to train deep convolutional networks [8]. Imagenette is a ten-class subset of ImageNet intended for efficient experimentation and includes ten categories used in this study [9]. This experiment uses a balanced sample of 300 images, not a full Imagenette evaluation.

3. Research Questions and Contribution

- RQ1: How does cartoonization method affect the top-1 recognition accuracy of a fixed pretrained ResNet18 classifier?
- RQ2: How do the methods perform under Gaussian blur, Gaussian noise, low illumination, and JPEG compression?
- RQ3: How do SSIM and PSNR relate to downstream recognition in this setup?
- RQ4: Which categories and examples show the largest recognition failures after cartoonization?

The defensible contribution at this stage is an empirical, task-aware comparison of three distinct cartoonization pipelines under a common classifier and selected image degradations, complemented by per-class and failure-case analyses. Do not claim this is the first such study or a confirmed literature gap until a broader, documented search checks directly comparable prior work.

4. Materials and Methods

4.1 Dataset and sample selection

A balanced subset of 300 images from ten Imagenette categories was used, with 30 selected images per class and a recorded selection seed of 42. The sample manifest is selected_300_images.csv. The final paper should specify the dataset release, image inclusion criteria, dimensions, license conditions, and handling of duplicate or corrupt files. The dataset source and release should be identified precisely in the final submission. Although the fast.ai Imagenette repository contains an Apache-2.0 license file, this should not be interpreted as independently licensing the underlying ImageNet images; the applicable ImageNet terms and any redistribution restrictions must be checked. The image files are therefore not bundled with this repository.

4.2 Cartoonization pipelines

Three transformations were evaluated: (i) an OpenCV-based cartoonization pipeline, (ii) pretrained AnimeGANv2, and (iii) pretrained Stable Diffusion image-to-image stylization. No model training was reported. Reproducibility requires the exact OpenCV operations and parameters, AnimeGANv2 checkpoint/source, Stable Diffusion model identifier/revision, prompts, inference settings, seeds, preprocessing, and output postprocessing.

4.3 Recognition evaluation

A pretrained ResNet18 classifier was applied to original and transformed images. Top-1 accuracy was calculated as correct predictions divided by valid evaluated samples. Six black Stable Diffusion outputs were excluded from the clean-image Stable Diffusion denominator, leaving 294 evaluated images; the original, OpenCV, and AnimeGANv2 sets each had 300.

Four conditions were evaluated: Gaussian blur (kernel 9×9, sigma 2), Gaussian noise (standard deviation 15, seed 42), low illumination (alpha 0.5), and JPEG compression (quality 30). For each condition, recognition metrics were recalculated on the intersection of image IDs with valid prediction records across all four methods. The matched sample sizes were 292 for blur, 297 for Gaussian noise, 295 for low illumination, and 294 for JPEG compression. Original outputs are preserved; corrected tables are reported below.

Four conditions were evaluated: Gaussian blur (kernel 9×9, sigma 2), Gaussian noise (standard deviation 15, seed 42), low illumination (alpha 0.5), and JPEG compression (quality 30). Blur, illumination, and JPEG summaries use matched subsets that exclude images unavailable in one or more method outputs. The Gaussian-noise summary currently uses a different matching/denominator procedure and should not be compared as though all methods used identical image IDs.

4.5 Image quality and consistency

Mean SSIM and PSNR were calculated between original images and clean cartoonized outputs. Six black Stable Diffusion outputs were excluded, leaving 294 valid images. Prediction consistency is the proportion of matched samples for which the classifier predicts the same class for original and transformed images; consistency is distinct from correctness.

Wilson 95% confidence intervals were calculated for each matched-sample accuracy. Pairwise comparisons within each degradation condition used exact McNemar tests on paired predictions, with Holm correction across the six method pairs for that condition. The clean-image statistical analysis remains subject to a separate audit; p-values rounded to 0.000000 in console output should be reported as thresholds (for example, p < 0.001), not as zero.

For clean-image accuracy, Wilson 95% confidence intervals were calculated on the common matched subset of 294 image IDs. Exact paired McNemar tests were performed for all six method pairs, with Holm adjustment across the six comparisons. All six clean-image comparisons remained significant after Holm correction (adjusted p-values ranged from 7.39×10^{-7} to 1.59×10^{-33}; report exact values in the accompanying results file). Degradation comparisons were likewise evaluated on matched image IDs within each condition, with exact paired McNemar tests and Holm adjustment within each condition.

5. Results

5.1 Clean-image recognition

Method	Matched samples	Correct	Accuracy
Original	294	237	80.61%
OpenCV	294	53	18.03%
AnimeGANv2	294	174	59.18%
Stable Diffusion	294	98	33.33%

On the common matched clean subset (n=294), original images achieved 80.61% accuracy (237/294). AnimeGANv2 achieved the highest accuracy among the cartoonization methods (59.18%; 174/294), followed by Stable Diffusion (33.33%; 98/294) and OpenCV (18.03%; 53/294). Relative to the matched original baseline, the absolute reductions were 21.43 percentage points for AnimeGANv2, 47.28 points for Stable Diffusion, and 62.59 points for OpenCV. These rankings are descriptive for this dataset and classifier.

5.2 Accuracy under degradations

Method	Gaussian blur (n=292)	Gaussian noise (n=297)	JPEG compression (n=294)	Low illumination (n=295)
Original	63.01%	72.39%	76.53%	75.93%
OpenCV	12.33%	13.47%	17.35%	16.61%
AnimeGANv2	40.07%	60.94%	57.82%	47.46%
Stable Diffusion	22.95%	24.24%	30.95%	23.39%

All four degradation comparisons below use identical image IDs across the four methods within each condition. The matched sample sizes differ by condition because Stable Diffusion has fewer valid prediction records: 292 for blur, 297 for Gaussian noise, 295 for low illumination, and 294 for JPEG compression.

AnimeGANv2 had the highest accuracy among the three cartoonization methods in each matched degradation condition. OpenCV had the lowest accuracy in each condition. The original degraded images achieved the highest accuracy overall. These results are descriptive of this sample, classifier, and the stated corruption settings.

Table 2 reports corrected matched-sample Top-1 accuracy. The reported sample size is common to all four methods within a condition, not across all conditions.

5.3 Image-quality metrics

Method	Valid images	Mean SSIM	Mean PSNR (dB)

AnimeGANv2	300	0.6564	18.5636
OpenCV	300	0.6469	17.6617
Stable Diffusion	294	0.4747	18.3549

AnimeGANv2 had the highest mean SSIM and PSNR in the clean-output comparison. Stable Diffusion had the lowest mean SSIM. These metrics measure similarity to the original, not aesthetic quality or semantic correctness.

5.4 Per-class and failure-case analysis

AnimeGANv2 had the highest reported accuracy in nine of ten classes; Stable Diffusion slightly exceeded AnimeGANv2 for garbage truck (60.00% versus 56.67%). Stable Diffusion per-class denominators vary because some outputs were invalid or unavailable. A gallery of 12 examples was generated where the original was correctly classified but at least one cartoonized version was misclassified. Confusion matrices used an 'Other classes' category for predictions outside the ten evaluated classes; corrected overall accuracy matched the main clean-image results.

5.5 Statistical comparisons

For each degradation condition, exact paired McNemar tests were conducted for all six method pairs and Holm-adjusted within that condition. All six pairwise comparisons were significant after Holm correction for Gaussian blur, Gaussian noise, and JPEG compression (adjusted $p < 0.001$). For low illumination, OpenCV versus Stable Diffusion had adjusted $p = 0.016858$, and the other five comparisons had adjusted $p < 0.001$. Clean-image paired tests were also independently recalculated on the common matched set of 294 image IDs; all six comparisons remained significant after Holm correction. Exact values are recorded in the accompanying clean-image and degradation results CSV files.

6. Discussion

Downstream recognition differed substantially by cartoonization pipeline. AnimeGANv2 retained more ResNet18 recognition accuracy than OpenCV and Stable Diffusion in this experiment. A plausible explanation is that transformations differ in how they preserve contours, texture, color, and class-discriminative structure, but accuracy alone does not establish this mechanism; failure examples may help investigate it.

SSIM/PSNR and recognition accuracy are complementary rather than interchangeable. A transformation may remain structurally similar while changing a feature important to the classifier, or differ substantially at the pixel level while preserving class identity. Across the four matched corruption settings, AnimeGANv2 remained the strongest of the cartoonization methods for this classifier. This does not establish that it is universally more robust across other datasets, classifiers, model versions, or corruption severities.

7. Limitations and Validity Threats

- Only 300 images from ten Imagenette classes were evaluated; results may not generalize to other datasets or real-world deployment.
- Only one classifier, ResNet18, was used.
- Valid sample counts differ: six clean Stable Diffusion outputs and three Stable Diffusion Gaussian-noise outputs were reported as black; per-class denominators vary.
- All four degradation conditions were recalculated on matched image IDs within each condition. Matched sample sizes differ across conditions (blur n=292, Gaussian noise n=297, low illumination n=295, and JPEG compression n=294), and missing Stable Diffusion records may introduce selection effects. Clean-image accuracy and paired statistical outputs were independently checked against the saved prediction records. Dataset provenance, corruption parameters, and full end-to-end reproducibility documentation remain necessary before publication.

- The Stable Diffusion checkpoint was stable-diffusion-v1-5/stable-diffusion-v1-5 at revision 451f4fe16113bff5a5d2269ed5ad43b0592e9a14. The implementation used StableDiffusionImg2ImgPipeline, FP16 weights, Safetensors, the PNDMScheduler, a CUDA Tesla T4 GPU, and attention slicing; the safety checker was enabled. Inputs were converted to RGB and resized to 512 × 512. The positive prompt was 'cartoon illustration, clean outlines, vibrant colors, stylized shapes, detailed background, high quality'; the negative prompt was 'blurry, low quality, distorted, deformed, text, watermark'. Strength was 0.45, guidance scale 7.5, and inference steps 20. Clean-image generation reset the seed to 42 for each image; the Gaussian-noise run used seed 42+i; the supplied blur, low-illumination, and JPEG cells did not explicitly pass a generator. Software versions: Python 3.13.15, PyTorch 2.11.0+cu130, Diffusers 0.40.0, Transformers 5.18.0, and Hugging Face Hub 1.33.0. Dataset provenance and corruption-generation parameters should be fully specified in the final methods section.
- SSIM and PSNR measure reference similarity, not aesthetic preference or semantic correctness.
- The clean and degradation paired tests, matched denominators, and Holm-adjusted p-values were checked against saved prediction records. Before submission, preserve the exact source manifests and code, verify all output-to-source mappings, document corruption severity/parameters, and ensure the reported SSIM/PSNR calculations and per-class tables reproduce from the archived outputs.
- The current literature review is preliminary, not systematic; related work on cartoon-domain recognition and image-domain robustness exists, so novelty must be framed as a specific controlled comparison and cannot yet be claimed as unprecedented.
- Comparable runtime and memory measurements were not available in the current record and are therefore not reported.

8. Conclusion

This study compared OpenCV cartoonization, AnimeGANv2, and Stable Diffusion image-to-image stylization using ResNet18 on a balanced 300-image Imagenette subset. On the common matched clean subset of 294 images, AnimeGANv2 achieved the highest recognition accuracy among the cartoonization methods (59.18%), followed by Stable Diffusion (33.33%) and OpenCV (18.03%); original images achieved 80.61%. AnimeGANv2 also achieved the highest accuracy among the cartoonization methods in all four evaluated degradation conditions. Six of 300 clean Stable Diffusion outputs were completely black, an output failure rate of 2.0%; its 33.33% recognition accuracy is conditional on the 294 valid outputs. Quality metrics and failure-case analysis provide complementary evidence that stylization should be evaluated using downstream tasks as well as visual similarity. Findings are specific to this sample, classifier, and pipeline settings. External validation, complete dataset/corruption documentation, reference verification, and a broader literature review remain necessary before submission or stronger novelty claims.

References

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Appendix A. Finalization checklist

- Save the executed Colab notebook (.ipynb) and an export of code cells.
- Record software versions, hardware/GPU, checkpoints, model revisions, prompts, seeds, preprocessing, and inference settings.
- Include selected_300_images.csv and document dataset source/version; verify ImageNet image terms separately from the fast.ai repository license, and do not redistribute source images unless permitted.
- Verify output-to-source mappings and row counts for all CSVs.
- Completed: recompute all four degradation comparisons on matched image IDs using the existing prediction CSVs; corrected outputs are saved in results/corrected_*_matched/.
- Completed: recompute clean-image and all four degradation comparisons on matched image IDs; independently verify paired McNemar tests and Holm adjustments; report p-values in scientific notation or as appropriate thresholds, never as literal zero.
- Document black-output exclusion rules and denominators.
- Add runtime/memory results only if measured comparably; otherwise state they were not evaluated.
- Complete a broader literature search and compare closest prior studies before claiming novelty.
- Check target journal formatting, references, dataset license, and permissions.